HomeLab Notes: visual style and layout guide

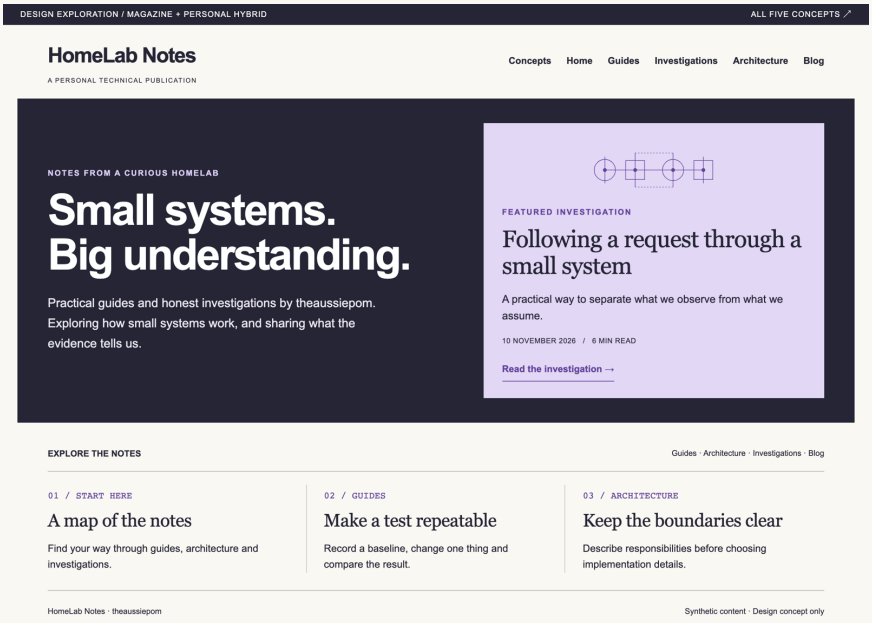
Version 1.2 | 10 November 2026 | Approved design direction: 04 + 05 hybrid

1. Direction and authority

The chosen direction combines the Technical magazine's confident homepage with the Quiet personal publication's warm, comfortable reading experience. Use this guide for future layout, typography, colour and component work. The Markdown is the canonical specification; the accompanying PDF is its printable edition.

The approved direction is implemented in the Astro/Starlight theme. These rules describe its shared visual language and the intentional adaptations to the native publishing shell. Reference images use synthetic content. Article wording, diagrams and dates in them are examples, not editorial requirements.

Keep the bold sans-serif masthead, dark opening panel and lilac featured article. Use warm paper surfaces and serif text for article summaries and extended reading. Carry purple accents consistently from discovery into the article. The homepage may be expressive; the article should give the text room to breathe.



Approved hybrid homepage reference

The reference image shows the visual intent. The numeric rules below take precedence where they deliberately improve the prototype, especially small labels, code size, tablet layout and mobile navigation. Dark mode is specified below. Editorial templates now apply this guide; final branding and real article content remain separate decisions.

2. Typography

Use local system fonts. No external font service or font download is required. Test fallback rendering because font availability and metrics vary by operating system.

Role	Font stack	Weight
Brand, display headings, navigation, labels	Arial, Helvetica, sans-serif	700 for emphasis; 400 for supporting text
Article body, lede, section headings, card titles	Georgia, 'Times New Roman', serif	400; 700 for inline emphasis
Code, commands, identifiers	ui-monospace, SFMono-Regular, Consolas, 'Liberation Mono', monospace	400

Type specimens

Small systems. Big understanding.

A practical way to understand the system.

```
curl --head https://service.example.test
```

Use genuine available weights. Arial commonly provides regular and bold; use 700 rather than relying on a synthetic 800 weight. Keep the expressive homepage headline in sans-serif and the featured article's title in serif.

Element	Desktop size / line height	Mobile size / line height
Masthead	29px / 1.15	25px / 1.15
Homepage headline	60px / 1.03	36-45px / 1.03
Article title	48px / 1.08	32px / 1.08
Featured article title	33px / 1.15	29px / 1.15
Article lede	23px / 1.45	22px / 1.50
Article body	18px / 1.70	18px / 1.70
Article H2 / H3	29px / 1.25; 23px / 1.30	29px / native heading rhythm; 23px
Discovery card title	25px / 1.20	25px / 1.20
UI and navigation	14-16px / 1.50	14-16px / 1.50
Labels and metadata	12px / 1.50 minimum	12px / 1.50 minimum
Code / table text	14px / 1.70; 16px / 1.50	Same sizes; contain overflow locally

Implement sizes in rem, based on the user's default font size; do not lock the root font size. Use fluid sizing between these anchors. Guides, architecture articles and investigations display their description as a lede. Blog posts retain their native metadata and article body. Preserve the native heading spacing and code line-height where the framework supplies them. Keep body text left-aligned and avoid justified columns. Target 60-75 characters per line, with the article width capped at 700px.

Use at most -0.04em tracking for display headings and -0.02em for serif titles. Body text uses normal tracking. Short uppercase labels may use 0.10em tracking; never use uppercase for paragraphs. Give paragraphs 1em separation and H2 sections about 2em above / 0.6em below. Essential labels and commands must remain legible at 200% zoom.

3. Colour system

Use semantic roles rather than scattering literal colours through component CSS. The light theme uses warm paper and a dark homepage panel. The dark theme uses an ink reading surface and muted violet panels. The hero retains its fixed dark surface in both modes.

Token	Value	Intended use
--hl-paper	#FAF8F3	Page and article background
--hl-ink	#252535	Primary text (hero uses fixed ink)
--hl-accent	#62429A	Links, labels, active navigation and focus
--hl-feature	#E2D7F5	Featured article background
--hl-on-dark	#FFFFFF	Display headings on dark surfaces
--hl-inverse-body	#EEEAF3	Body text on dark surfaces
--hl-inverse-accent	#DFD0FA	Links and labels on dark surfaces
--hl-callout	#F0EBF5	Informational callout background
--hl-rule	#BEB9B1	Decorative section separators
--hl-rule-soft	#D0CBC3	Decorative column dividers
--hl-diagram-paper	#F4F0E9	Diagram node background

Use ink for body text and the framework's contrast-tested grey scale for supporting UI; establish hierarchy with size, weight and spacing instead of low-opacity text. The two rule colours are decorative only. Use the accent or ink for a boundary that is necessary to recognise an input, button or selected state.

Measured text contrast

These ratios are calculated from the listed solid sRGB colours, rounded to two decimals. They validate colour pairs, not the accessibility of the finished website. Require at least 4.5:1 for normal text and 3:1 for essential control boundaries and indicators.

Pair	Contrast
Ink on paper	14.19:1
Accent on paper	7.16:1
Ink on lilac	10.95:1
Accent on lilac	5.52:1
White on ink	15.06:1
Inverse body on ink	12.69:1
Accent on callout	6.48:1

Underline links within prose. Pair status colours with text or an icon and a label. Use a 3px accent focus outline with a 3px offset on light surfaces, and the inverse accent on dark surfaces. Verify the actual adjacent colours during implementation. Success, warning and error palettes are not approved here; preserve accessible framework treatments until separately reviewed.

Dark-mode reference

The following semantic overrides belong to the same hybrid design. Keep display hierarchy, spacing and typography identical across modes. Do not use a blanket colour inversion.

Token	Value	Intended use
--hl-paper	#1B1B27	Page and article background
--hl-ink	#F1EEE8	Primary reading text
--hl-accent	#CBB5F1	Links, active controls and focus
--hl-feature	#332B46	Featured article panel
--hl-callout	#2A2539	Notes, diagrams and inline code
--hl-rule	#615A6D	Decorative section separators
--hl-rule-soft	#484252	Decorative dividers

The fixed hero remains #252535 with white headings, #EEEAF3 body text and #DFD0FA labels. The feature card uses the theme-dependent panel, text and accent roles. Native search, sidebar, contents, menus and code blocks follow the same selected theme. Warning, danger and tip treatments retain their native semantic colours.

Light, Dark and Auto must remain usable on desktop and mobile. In Auto, changing the operating system preference updates the page. Explicit choices persist through navigation and reloads. Keep focus indicators visible on both page and panel surfaces. Inspect syntax highlighting and controls after framework updates.

4. Homepage layout and responsive rules

Desktop: 1100px and wider

Use a centred homepage content wrapper with a maximum width of 1160px. Native Starlight outer gutters surround it; the hero adds 42px internal side padding. The header is at least 102px tall and may grow with content. Place the masthead at left, native search in the middle and theme/social controls at right. Put the five topic links in a wrapping row above the hero. Preserve access to Start Here, Guides, Architecture, Investigations, Blog and search; the prototype's reduced navigation is not a reason to drop production features.

The hero spans the wrapper's full width. Use two columns with a 1.18:1 ratio, a 46px gap and 34px vertical padding. Align the introduction and featured article near their vertical centres. The left column contains a short context label, a strong headline and a concise personal introduction. Limit the introduction to about 455px. Avoid fixed heights or manual line breaks that fail at intermediate widths.

The lilac feature uses 24-28px padding, one representative diagram or image, a serif article title, a short summary, metadata and a clear reading link. Its image should support the subject. Abstract graphics in the reference are placeholders, not a permanent brand asset.

Below the hero, show an understated section label and a three-column discovery row. Use 24-32px gaps or internal padding, serif card titles and short summaries. Avoid enclosing every item in a heavy card. Surface topic categories and a route to the archive as content grows.

Tablet: 701-1099px

Use the native outer gutters and 28px internal hero padding. The native header changes at 800px; keep the homepage theme control visible below that width. Stack the hero by default; a two-column hero is permitted only if both columns remain readable and test content fits. Discovery rows may use two columns, with the remaining item taking a natural next row. Do not shrink text to preserve a desktop composition.

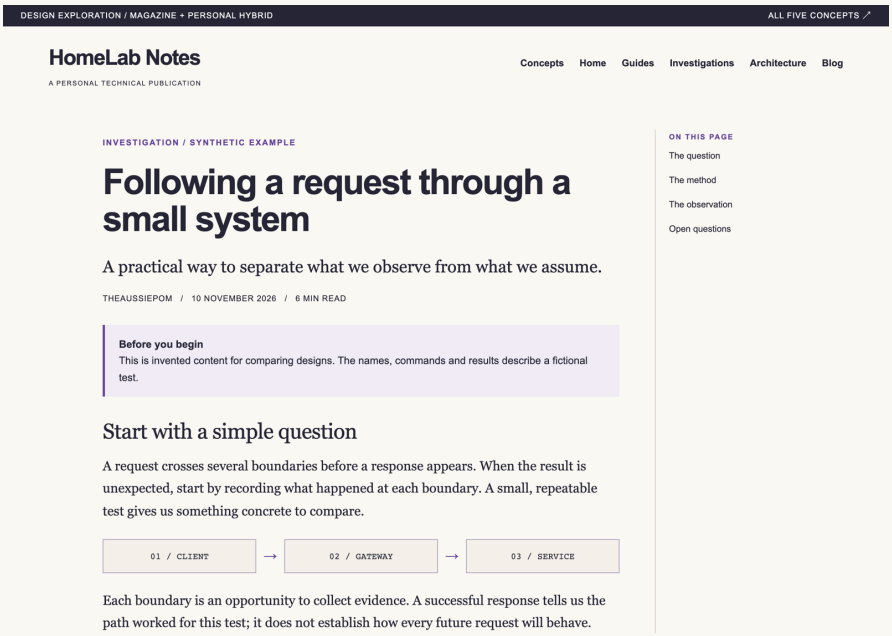
Mobile: up to 700px

Use native outer gutters and 22px internal hero side padding. Stack the hero and discovery content. Use 28px hero vertical padding, 20px feature padding and about 26px between the introduction and feature. The homepage has visible wrapping topic links, search and a theme selector, so it needs no menu drawer. Documentation and blog pages retain the native labelled Menu button and sidebar. On mobile the homepage header is 132px tall to accommodate the theme selector.

Shared spacing and geometry

Use a 4px spacing scale: 4, 8, 12, 16, 20, 24, 28, 32, 40, 48 and 64px. The approved 22px gutter, 34px hero padding and 46px hero gap are documented composition exceptions. Use square or subtly rounded corners, 0-4px. Use thin rules and colour blocks for structure; shadows are not the default. Prefer content-driven heights.

5. Article layout



Approved hybrid article reference

Cap the reading column at 700px inside Starlight's native article shell. Retain its 240px section sidebar from 800px and right-hand contents column from 1152px; below those widths use its native collapsible navigation and contents controls. These production breakpoints deliberately replace the prototype's 930px region and 1100px contents breakpoint, preserving keyboard and responsive behaviour.

For guides, architecture articles and investigations, use the native title followed by a description lede and a compact metadata row: type, author, publication date, optional update date and topics. A local draft notice appears after the title when needed. Blog pages retain the plugin's native author, date, reading metrics and tags; do not duplicate that metadata. Landing pages have no article introduction. Use a single H1 followed by a logical heading hierarchy. Keep paragraphs and diagrams in the reading column. Wide tables or code may scroll within their own container, never force horizontal scrolling of the page.

Use sans-serif for the title, metadata and navigation; serif for the lede, body and section headings. The desktop title is 48px, the body is 18px with 1.7 line height. Mobile retains the body size. The table of contents may be sticky if it stays inside the viewport and does not obscure focus or headings.

The reference screenshot is a top-of-article excerpt. Real articles must preserve useful navigation through the full page, including previous/next or related content where appropriate. Long titles, multiple authors, absent imagery and long code lines must all fit without changing the design's basic rules.

6. Components and interaction

Component	Required treatment
Links	Accent on light surfaces; inverse accent on dark. Underline prose links. Hover may strengthen the underline; focus must remain explicit.
Navigation	Clear active item using text weight plus an indicator. Preserve visible labels and access to search. Do not rely on colour alone.
Mobile menu	Labelled button with expanded state. Keyboard operable; keep focus order logical. Restore focus on close when using a modal drawer.
Featured article	Lilac surface, serif title, concise summary and one primary reading action. Avoid nested interactive targets.
Informational callout	Callout surface, 3px accent left edge, clear text label, 16px body and 16-20px padding. Do not use this style for every severity.
Code block	Native light/dark syntax themes, 14px monospace and framework line height. Preserve command whitespace; offer local scrolling for long lines.
Inline code	Monospace, restrained tinted surface, enough padding to distinguish it without breaking line rhythm.
Table	16px text, strong column headers, generous cell padding and light row rules. Associate headers semantically. Scroll only the table when needed.
Diagram	Callout-tinted nodes, labelled connections and readable text. Provide a meaningful text alternative. Distinguish states with shape or labels as well as colour.
Search	Retain Pagefind/Starlight search behaviour, keyboard access, result feedback and empty states. Style without replacing its semantics.

Size primary touch targets to at least 44px by 44px where practical, including Menu, search and copy buttons. Avoid hover-only information. Respect reduced-motion preferences; transitions should be brief and functional, with no automatic animation required for reading.

Preserve framework code-copy controls and accessible labels. Syntax-highlighting colours require their own contrast check against the code background. Do not wrap copyable shell commands in ways that alter their content; the prototype's display wrapping is not a production requirement.

Use authentic article diagrams, photographs or illustrations only when they contribute to understanding. Keep imagery responsive and reserve its layout space to avoid jumps. Sanitise all technical examples and screenshots under the existing editorial policy. No private infrastructure detail belongs in design fixtures.

7. Astro and Starlight implementation

Apply the guide through shared tokens, scoped styles and the smallest necessary components. Keep Markdown/MDX content, Starlight navigation, headings, search, code rendering, RSS, sitemap and starlight-blog integration intact. The static publishing architecture and deployment pipeline remain authoritative.

Start with theme tokens and typography, then the custom homepage and shared article treatment. Map semantic colours to the installed Starlight version's supported theme variables. Use documented component extension points; verify current APIs before implementation. Scope homepage composition separately from article rules so the dark hero does not leak into documentation pages.

Core token example

```
:root {
  --hl-paper: #faf8f3;
  --hl-ink: #252535;
  --hl-accent: #62429a;
  --hl-feature: #e2d7f5;
  --hl-callout: #f0ebf5;
  --hl-font-ui: Arial, Helvetica, sans-serif;
  --hl-font-reading: Georgia, 'Times New Roman', serif;
  --hl-font-code:
    ui-monospace, SFMono-Regular, Consolas, 'Liberation Mono', monospace;
  --hl-reading-size: 1.125rem;
  --hl-reading-line-height: 1.7;
}
```

This abbreviated light palette illustrates the roles. The authoritative mappings and both palettes are in `src/styles/hybrid.css`; its base tokens are dark and its `[data-theme="light"]` override is light. Native Starlight controls provide Light, Dark and Auto, persist an explicit choice, and follow system preference in Auto. No additional theme storage or custom switching script is required.

`HomeHero.astro` supplies the homepage and falls back to the native hero elsewhere. `SiteTitle.astro` supplies the masthead. `ArticleTitle.astro` preserves the native title, adds the article lede and metadata, and labels local drafts. Keep its metadata at 14px with wrapping rows.

Use the editorial workflow and templates (../editorial/workflow.md) for article structure. They use native Starlight notes and steps, Markdown tables and code. The Australian English UI dictionary supplies blog labels because plugin `en` strings do not fall back to `en-AU`; maintain it when blog controls change.

Implementation acceptance

- Compare the rendered homepage and article with the approved references at 1200px and 390px.
- Check 320px, 700px, 768px, 1024px and 1440px widths, long titles and 200% zoom.
- Verify menu, search, article anchors, focus order, visible focus, copy controls and theme switching with keyboard input.
- Check actual text, control and syntax colours; run automated accessibility checks and inspect the result manually.
- Check article prose, lists, tables, code, diagrams, callouts and image captions with realistic synthetic content.
- Run the repository's required validation and security checks before integration. Verify preview rendering before production deployment.

8. Maintenance and verification record

What is established

The user selected the 04 + 05 hybrid on 10 November 2026. The reference prototypes were rendered at 1200px and 390px; both homepage and article had no horizontal page overflow, their mobile menus opened, and local links resolved. The reference PDF was visually inspected. This guide's solid-colour contrast ratios were calculated from the listed values.

Implementation evidence and limits

The hybrid uses shared CSS and overrides for the homepage, masthead and article introduction. Validation covers formatting, lint, types, production builds, generated links and Chromium tests. Both palettes receive axe checks; responsive checks cover 320-1440px. Tests exercise navigation, search, theme persistence, system preference, enlarged text, RSS and sitemap. See the acceptance record for deployment evidence.

Automated checks are not a complete accessibility audit, and root-text enlargement does not establish browser-zoom equivalence. Firefox, Safari, screen readers and real article content remain unverified. Templates and synthetic workflow fixtures exercise presentation, not the truth of future technical claims.

Keeping the specification current

Edit this Markdown first. Update the PDF in the same change, including its version and date, and render every PDF page for visual inspection. Update reference images when an approved visual change makes them stale. Keep images synthetic and public-safe. Do not change the deployed design merely to make it match an outdated screenshot; resolve the specification deliberately.

Use pull requests for integration and follow the existing validation gates. Changes to typography roles, primary palette, homepage composition or article reading width require an explicit design decision. Routine implementation adjustments may follow the documented responsive and accessibility rules.

Design review questions

- Does the homepage retain the confidence and visual character of option 04?
- Does the article preserve the warmth and reading comfort of option 05?
- Can readers find guides, investigations, architecture and blog content quickly?
- Does the same hierarchy survive a narrow screen, long title and enlarged text?
- Are framework capabilities and public-content boundaries preserved?

Document map

- Printable style guide (style-guide.pdf)
- Repository README (../../README.md)
- Architecture decision (../decisions/0001-static-publishing.md)
- Editorial workflow and templates (../editorial/workflow.md)
- Editorial sanitisation policy (../editorial/sanitisation.md)
- Foundation acceptance evidence (../acceptance.md)